

AthDGC: An Open Diachronic Greek Treebank with Indo-European Parallels

A formal short data paper for journal submission

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AthDGC (“Athens-PROIEL”) is an open, end-to-end pipeline and dataset that provides the first continuously updated, dependency-parsed treebank of the entire Greek language — from Homeric and Archaic texts to Modern Greek — with cross-lingual alignment to Latin (Vulgate), Gothic (Wulfila), Old Church Slavonic (Marianus), and Classical Armenian via the verse-level cross-alignment of the New Testament. AthDGC follows the PROIEL XML 2.0 schema, uses the Stanford Stanza PROIEL-trained pipeline for annotation, and applies LaBSE sentence embeddings and multilingual-BERT attention for word-level alignment. As of v0.4, the corpus contains 89.9M total rows, 4.08M annotated Greek rows, and 6,861 NT-aligned Greek verses cross-aligned to four sister IE witnesses.

1. Overview

1.1 Context and motivation

Greek is the longest continuously attested member of the Indo-European family. The PROIEL Treebank (Haug and Jøhndal 2008; Eckhoff et al. 2018) established the gold standard for syntactically annotated parallel Indo-European Bible corpora, but its scope is restricted to the New Testament and to a small number of language witnesses. No openly licensed, end-to-end pipeline currently covers the full diachronic record of Greek at the same level of annotation, with explicit argument-structure tagging, and with reproducible cross-lingual alignment to its sister IE languages. AthDGC closes that gap.

The project’s specific scholarly focus is *retranslation*: the same canonical text — the Iliad, the New Testament, the Septuagint Psalms, classical historiography — is re-rendered into Greek across periods (Homeric → Koine → Byzantine → Modern) and into sister languages (Latin, Gothic, OCS, Armenian) across the Indo-European family.

1.2 Repository location

- Source code: <https://github.com/AthDGC/Diachronic-Linguistics-Platform>
- Public showcase: <https://athdgc.github.io>
- Zenodo deposition: <https://doi.org/10.5281/zenodo.20171725>

1.3 Institutional context

AthDGC is developed at the **National and Kapodistrian University of Athens (EKPA)**, Division of Language-Linguistics, Department of English Language and Literature, School of Philosophy, under the direction of Prof. Nikolaos Lavidas. Compute is supplied by GRNET ARIS, allocation pa260305.

2. Method

2.1 Discovery

Each day, the discovery stage probes open-access repositories — archive.org, Perseus Digital Library, First1K Greek, Wikisource, the Diorisis Corpus, OpenGreekAndLatin/PerseusDL — for new or updated Greek and parallel-language source material.

2.2 Filtering

Candidate texts are filtered by Greek-script ratio (at least 75% of alphabetic characters in the Greek Unicode blocks), a Path-B line filter for bilingual editorial apparatus, and content-hash deduplication.

2.3 Conversion

Surviving text is converted into the PROIEL XML 2.0 schema with sentence-level structure.

2.4 Annotation

Annotation is performed sentence by sentence with the Stanford Stanza pipeline (Qi et al. 2020) using the PROIEL-trained model `grc_proiel`. Analogous PROIEL-trained Stanza models are used for Latin (`la_proiel`), Old Church Slavonic (`cu_proiel`), and Gothic (`got_proiel`); Classical Armenian is annotated through a separate UD-style pipeline currently under development.

2.5 Argument-structure capture

Beyond standard dependency annotation, AthDGC extracts an explicit argument-structure frame for every verb token: the subject (`nsubj` / `nsubj:pass`), direct object (`obj`), indirect / oblique arguments (`iobj`, `obl:arg`), voice (active / middle / passive), and aspect (perfective / imperfective).

2.6 Cross-lingual alignment

Sentence-level alignment uses LaBSE embeddings (Feng et al. 2022); word-level alignment uses multilingual-BERT attention via the AwesomeAlign procedure (Dou and Neubig 2021). Phonetic cognate scoring uses ASJP sound-class encoding and LingPy edit distance (List 2014).

3. Dataset description

Field	Value
Object name	AthDGC corpus, v0.4 release
Format	PROIEL XML 2.0 (primary); JSONL partitions
Creation dates	2025-09 onwards; v0.4 = 2026-05
Dataset creator	Lavidas, N. and the Athens Digital Glossa Chronos Research Network
Language	grc, gkm, ell + parallels in lat, got, chu, xcl
Licence	Code Apache-2.0; metadata + alignments CC-BY-4.0; per-source raw text under its original licence
Repository	Zenodo, GitHub, Hugging Face
Publication date	2026-05-27 (v0.4)

4. Reuse potential

AthDGC supports four primary reuse classes: historical syntactic research, Indo-European comparative syntax, computational diachronic NLP, and digital editions / pedagogy.

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Research team

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